

0. Definitions

$$h[k] = h_d[k] - \delta h[k], \quad \Delta h_d[k] = h_d[k+1] - h_d[k]$$

$$a'_{hd}[k] := \left. \frac{da_h}{dh} \right|_{h_d[k]}, \quad a_r[k] := a_h[k] - a_h(h_d[k])$$

$$h_d(t) = h_0 + A_h \sin(2\pi f t)$$

$$a_{var} = a_{cov} = 0.05, \quad T_s = 1/1600, \quad \beta = 1 - T_s/\tau, \quad \tau = 0.35 \text{ s} \Rightarrow \beta = 0.99821$$

1. True

$$a_h[k+1] \approx a_h[k] + a'_{hd}[k] \Delta h_d[k] + (1 - \lambda_c) a'_{hd}[k] \delta h[k] + a'_{hd}[k] F_{dh}[k] e_{a_h}[k] + a'_{hd}[k] \varepsilon_h[k]$$

$$a'_{hd}[k+1] \approx a'_{hd}[k], \quad a_r[k] = -a'_{hd}[k] \delta h[k]$$

$$a'_{hdm}[k] := +\sqrt{\text{Var}(a_r[k]) / \text{Var}(\delta h[k])} = a'_{hd}[k] \quad (a'_{hd} > 0)$$

$$a'_{hd}[k+1] = \beta a'_{hd}[k] + (1 - \beta) a'_{hdm}[k], \quad a'_{hdm} \equiv a'_{hd}$$

2. $\hat{a}'_{hdm}[k]$

$$\bar{a}_h[k+1] = (1 - a_{var}) \bar{a}_h[k] + a_{var} \hat{a}_h[k+1], \quad \hat{a}_r[k] = \hat{a}_h[k] - \bar{a}_h[k]$$

$$\hat{\sigma}_{\hat{a}_r}^2[k+1] = (1 - a_{cov}) \hat{\sigma}_{\hat{a}_r}^2[k] + a_{cov} \hat{a}_r^2[k+1]$$

$$\overline{\delta \hat{h}}[k+1] = (1 - a_{var}) \overline{\delta \hat{h}}[k] + a_{var} \delta \hat{h}_3[k+1], \quad \delta \hat{h}_r[k] = \delta \hat{h}_3[k] - \overline{\delta \hat{h}}[k]$$

$$\hat{\sigma}_{\delta \hat{h}}^2[k+1] = (1 - a_{cov}) \hat{\sigma}_{\delta \hat{h}}^2[k] + a_{cov} \delta \hat{h}_r^2[k+1]$$

$$\hat{a}'_{hdm}[k] := +\sqrt{\hat{\sigma}_{\hat{a}_r}^2[k] / \hat{\sigma}_{\delta \hat{h}}^2[k]}$$

3. Transfer under $\Delta h_d \neq 0$ and $E[\hat{a}'_{hdm}]$

$$a_h(h_d(t)) - a_h(h_0) \approx a'_{hd} A_h \sin(2\pi f t) \quad (\text{deterministic component of } \hat{a}_h)$$

$$\lambda_{leak} \approx \frac{2\pi f T_s}{a_{var}} \quad (2\pi f T_s \ll a_{var} \ll 1)$$

leaked sinusoid in $\hat{a}_r$: amplitude $\lambda_{leak} a'_{hd} A_h$, power $\lambda_{leak}^2 a'_{hd}{}^2 A_h^2/2$

$$\mathbb{E}[\hat{\sigma}_{\hat{a}_r}^2] = \underbrace{\hat{a}'_{hd}{}^2 \sigma_{\delta h}^2}_{\text{self}} + \underbrace{\sigma_u^2}_{\text{corrections}} + \underbrace{\lambda_{leak}^2 a'_{hd}{}^2 A_h^2/2}_{\text{leak}}, \quad \mathbb{E}[\hat{\sigma}_{\delta h}^2] = \sigma_{\delta h}^2$$

$$\mathbb{E}[\hat{a}'_{hdm}]^2 \approx \frac{\hat{a}'_{hd}{}^2 \sigma_{\delta h}^2 + \sigma_u^2 + \lambda_{leak}^2 a'_{hd}{}^2 A_h^2/2}{\sigma_{\delta h}^2}$$

$$\Delta h_d = 0 : \mathbb{E}[\hat{a}'_{hdm}] \rightarrow \sqrt{\hat{a}'_{hd}{}^2 + \sigma_u^2/\sigma_{\delta h}^2}; \quad \hat{a}'_{hd} \rightarrow 0 : \mathbb{E}[\hat{a}'_{hdm}] \rightarrow \sigma_u/\sigma_{\delta h}$$

$$\text{leak-dominated: } \mathbb{E}[\hat{a}'_{hdm}] \rightarrow \frac{\lambda_{leak} A_h}{\sqrt{2} \sigma_{\delta h}} a'_{hd}$$

4. Esti

$$\begin{aligned} \hat{a}_h[k+1] &= \hat{a}_h[k] + \hat{a}'_{hd}[k] \Delta h_d[k] + (1 - \lambda_c) \hat{a}'_{hd}[k] \delta \hat{h}[k] \\ \hat{a}'_{hd}[k+1] &= \beta \hat{a}'_{hd}[k] + (1 - \beta) \hat{a}'_{hdm}[k] \end{aligned}$$

5. Error

$$\hat{a}'_{hdm}[k] = \hat{a}'_{hd}[k] + \kappa[k] e_{a'_h}[k] + n_{var}[k], \quad \mathbb{E}[n_{var}] = \mathbb{E}[\hat{a}'_{hdm}] - \hat{a}'_{hd} \quad (\S 3)$$

$$\Delta h_d = 0 : \mathbb{E}[n_{var}] \approx \frac{\sigma_u^2}{2\hat{a}'_{hd}\sigma_{\delta h}^2}; \quad \Delta h_d \neq 0 : \mathbb{E}[n_{var}] \approx \frac{\sigma_u^2 + \lambda_{leak}^2 a'_{hd}{}^2 A_h^2/2}{2\hat{a}'_{hd}\sigma_{\delta h}^2}$$

$$e_{a'_h}[k+1] = [1 - (1 - \beta) \kappa[k]] e_{a'_h}[k] - (1 - \beta) n_{var}[k]$$

$$\kappa > 0 : e_{a'_h}^{ss} = -\mathbb{E}[n_{var}]/\kappa; \quad \kappa = 0 : e_{a'_h}[k+1] = e_{a'_h}[k] - (1 - \beta) n_{var}[k] \quad (\text{integrator})$$

$$\beta \text{ sets speed/smoothing only: } \text{pole} = 1 - (1 - \beta) \kappa, \quad \kappa, \mathbb{E}[n_{var}] \text{ are } \beta\text{-invariant}$$

$$T_s/(1 - \beta) = \tau \text{ (response)}, \quad \text{Var}(\hat{a}'_{hd}) \approx \frac{1-\beta}{2} IF_{var} R_{var} \quad (\beta \rightarrow 1)$$

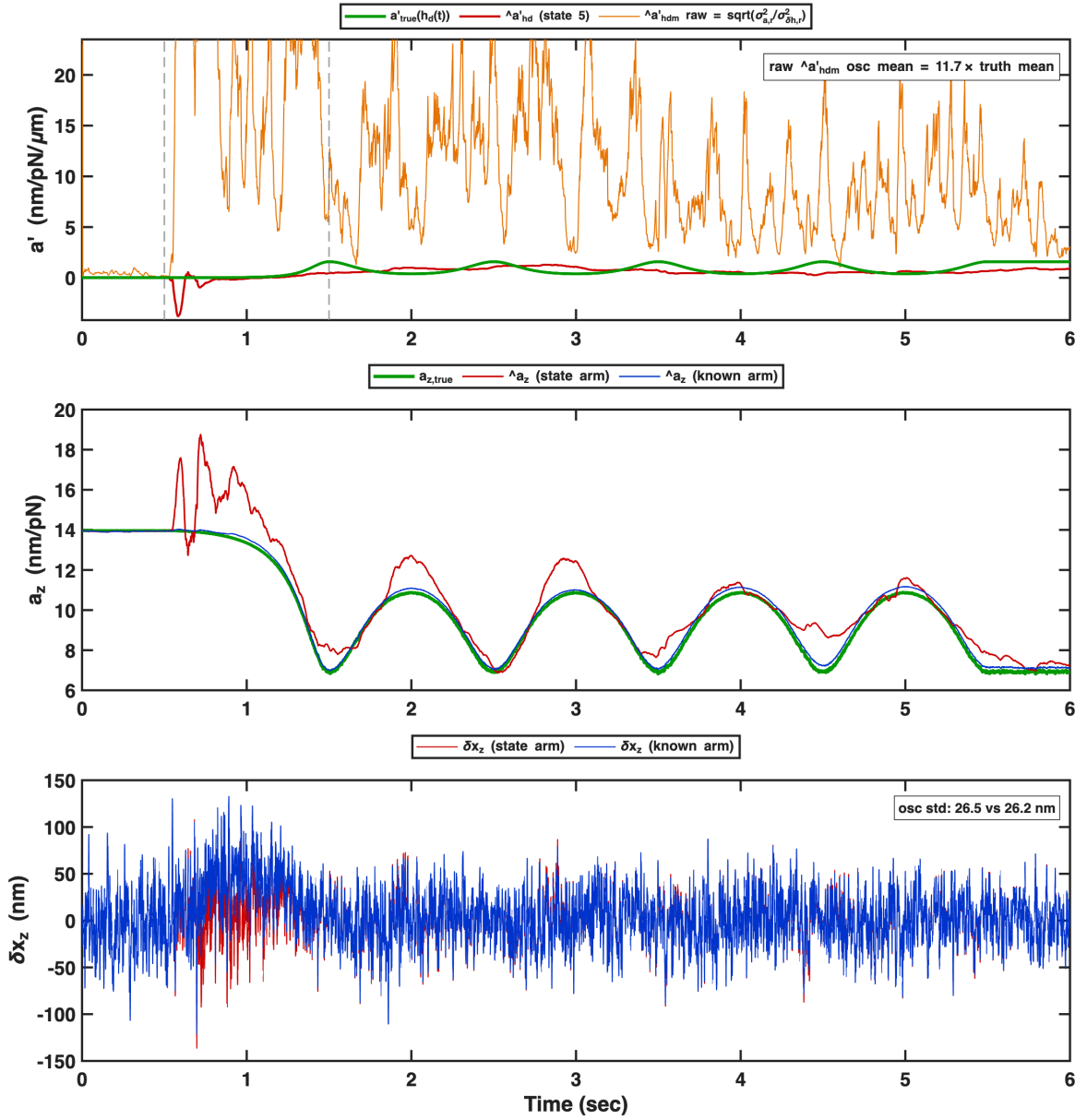
6. $\mathbf{F}_e$

$$x_e = [\delta h_1, \delta h_2, \delta h_3, e_{a_h}, e_{a'_h}]^T$$

$$\mathbf{F}_e[k] = \begin{bmatrix} 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & \lambda_c & -F_{dh}[k] & 0 \\ 0 & 0 & (1 - \lambda_c) \hat{a}'_{hd}[k] & 1 + \hat{a}'_{hd}[k] F_{dh}[k] & \Delta h_d[k] \\ 0 & 0 & 0 & 0 & 1 - (1 - \beta) \kappa[k] \end{bmatrix}$$

$$q[k] = [0, 0, -\varepsilon_h[k], \hat{a}'_{hd}[k] \varepsilon_h[k], -(1 - \beta) n_{var}[k]]^\top$$

7. Simulation (E1S, 1 Hz sweep, seed 1)



E1S (1 Hz sweep, seed 1): raw $\hat{a}'_{hdm}$ (orange) reads 11.7 $\times$ truth (leak formula $\approx 10\times$); the a' -as-state route (red, $F_{45} = \Delta h_d$, no var statistic) stays at truth scale; tracking unharmed (26.5 vs 26.2 nm).